
Microprocessor Systems and Interfacing

EEE 342

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Lecture # 08

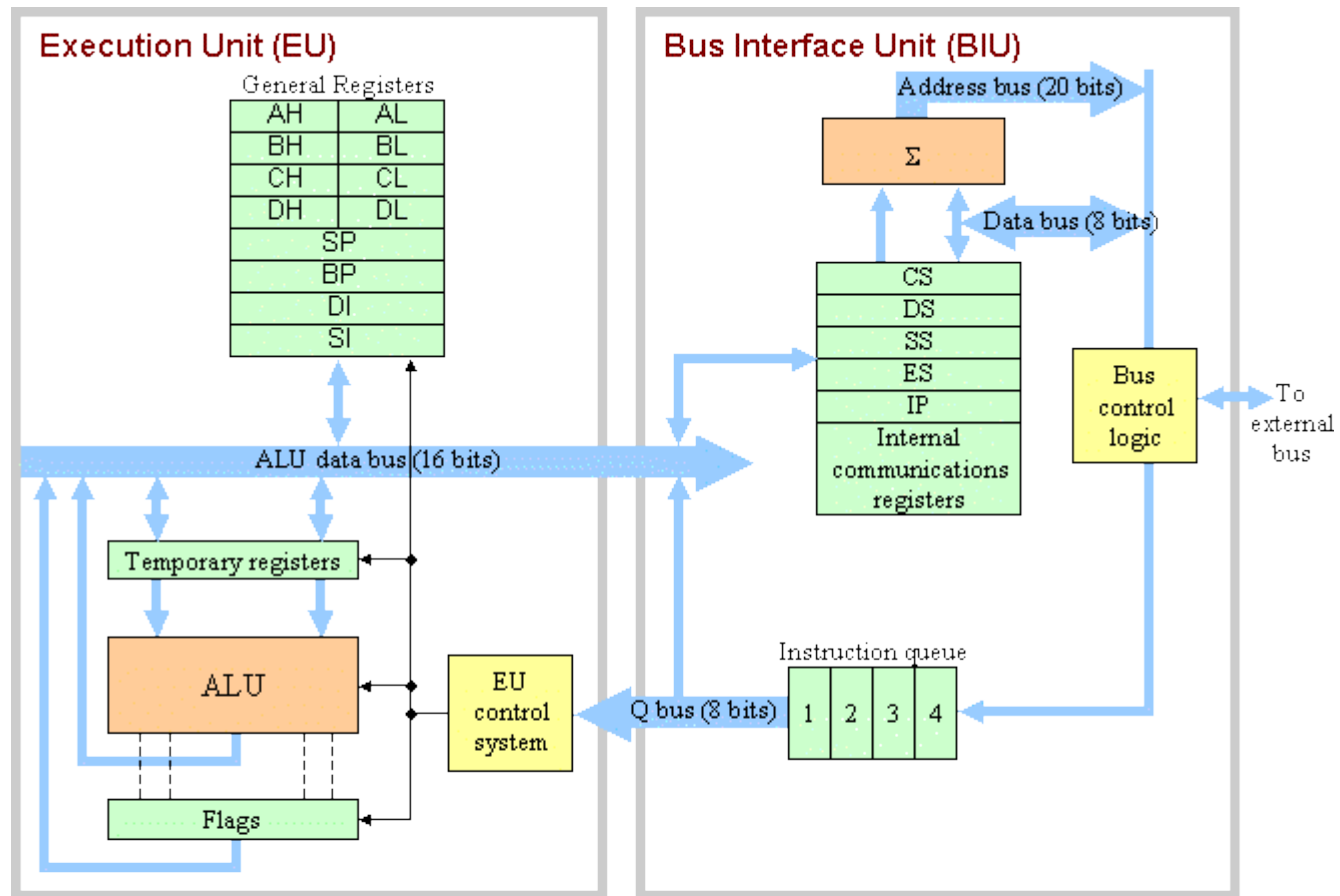
Week 5 Lecture 1

Microprocessor 8088

■ Outline

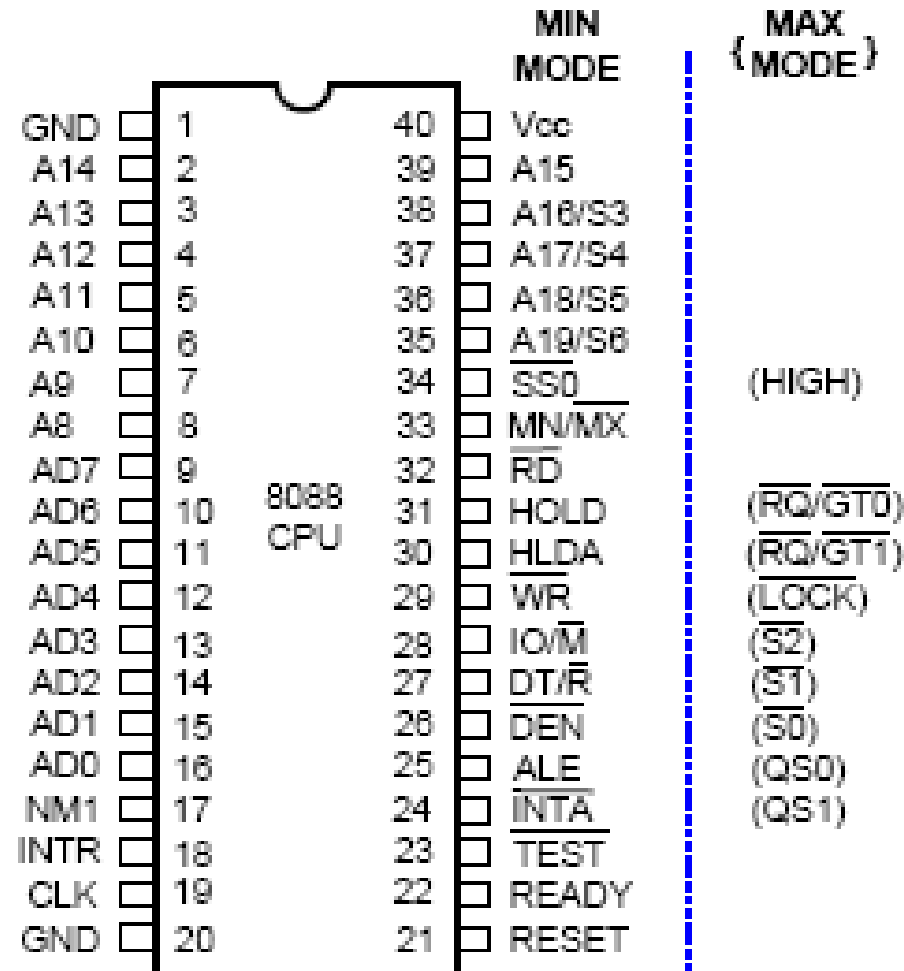
- ❑ Architecture of 8088
- ❑ Pin layout of 8088
- ❑ Pin Explanation of 8088
- ❑ Type of Memories
- ❑ Pin Configuration of Memory Chip
- ❑ Interfacing with 8088

Architecture



- Static registers (groups of D Flip-Flops) used to hold or transfer binary data
- Logic gate circuits designed to perform arithmetic or logical functions
- Logic gate circuits designed to provide internal control to processor
- Internal data busses used to pass information between components

Pinout of 8088



8088 pin diagram

Pinout of 8088

Pin Number

1

2-8

9-16

17

18

19

20

21

22

23

24

Description

GND - Ground

ADDR - Address Bus

Data - Address Data Bus

NMI - Non-Maskable Interrupt

INTR - Interrupt Request

CLK - Clock

GND - Ground

RESET - Reset

READY - Ready

TEST - Test (Active Low)

INTA - Interrupt Acknowledge
(Active Low)

Pinout of 8088

25	ALE - Address Latch Enable
26	DEN - Data Enable (Active Low)
27	DT/R - Data Transmit/Receive
28	IO/M - Status Line
29	WR - Write (Active Low)
30	HLDA - Hold A
31	HOLD - Hold
32	RD - Read (Active Low)
33	MN/MX - Minimum or Maximum Mode
34	SSO - Status Line (Active Low)
35-39	ADDRESS/STATUS - Address/Status Information
40	Vcc - Positive Power Supply

Pin Explanation

AD7-AD0 : address/data bus(multiplexed)

memory address or I/O port no : whenever ALE = 1

data : whenever ALE = 0

high-impedance state : during a hold acknowledge

AD15-AD8 : address/data bus(multiplexed)

memory address bits A15-A8 : whenever ALE = 1

data bits D15-D8 : whenever ALE = 0

high-impedance state : during a hold acknowledge

A19/S6-A16/S3 : address/status bus(multiplexed)

memory address A19-A16, status bits S6-S3

S6 : always remain a logic 0

S5 : indicate condition of IF flag bits

S4, S3 : show which segment is accessed during current bus cycle

S4, S3 : can used to address four separate 1M byte memory banks by decoding them as A21, A20

Pin Explanation

RD(bar) : read signal

data bus receive data from memory or I/O device :RD'=0

READY :

μ enter into wait states and remain idle : READY = 0

INTR : interrupt request

used to request a hardware interrupt

if INTR is held high when IF = 1 : μ enter interrupt acknowledge cycle(INTA' become active) after current instruction has complete execution

TEST(bar)(BUSY(bar)) : tested by the WAIT instruction

WAIT instruction function as a NOP : if TEST'= 0

WAIT instruction wait for TEST' to become 0:if TEST'=1

NMI : non-maskable interrupt

similar to INTR except that no check IF flag bit

if NMI is activated : use interrupt vector 2

RESET :

μ : reset if RESET held high for a minimum of four clock

Pin Explanation

CLK(CLOCK) : provide basic timing to μ duty cycle of 33%

VCC(power supply) : +5.0V, $\pm 10\%$

GND(Ground) : two pins labeled GND

MN/MX(bar) : select either minimum or maximum mode (HIGH for minimum mode)

BHE(bar)/S7 : bus high enable

status of S7 : always a logic 1

BHE=0 at least one byte of current transfer is to be made AD15-AD8 (HIGHER ORDER BYTE)

IO/M(bar)(8088) or M/IO(bar)(8086) : select memory or I/O address bus : whether memory or I/O port address

WR(bar) : write signal(high impedance state during hold ack).
strobe that indicate that output data to memory or I/O
during WR(bar)=0 : data bus contains valid data for M or I/O

Pin Explanation

S6 : always remain a logic 0

S5 : indicate condition of IF flag bits

S4, S3 : show which segment is accessed during current bus cycle

S4, S3 : can used to address four separate 1M byte memory banks by decoding them as A21, A20

S4	S3	Function
0	0	Extra Segment
0	1	Stack Segment
1	0	Code or no Segment
1	1	Data Segment

Memories

- Every microprocessor-based system has memories
- Semiconductor memories are used as primary storage of data and code
- These memories are directly connected to CPU
- Two main type of memories
 - RAM (Random Access Memory)
 - ROM (Read Only Memory)
- Memory Capacity
 - The number of bits a memory chip can store

Memories

- Some Important numbers to consider
 - ❑ 1K = 1024 Location → 10 Address Pins, 2^{10}
 - ❑ 2K = 2048 Location → 11 Address Pins 2^{11}
 - ❑ 4K = 4096 Location → 12 Address Pins 2^{12}
 - ❑ 1M = 1 Mega Location → 20 Address Pins
 - ❑ 400H represents 1K-byte of Memory
 - ❑ 1000H represents 4K
 - ❑ 2000H represents 8K

Memories

■ Memory Organization

- Each memory location contains 2^X locations, where x is the number of address pins on the chip
- Each memory location contains y bits, where y is number of data pins on the chip.
- Total bits = $2^X * Y$ bits

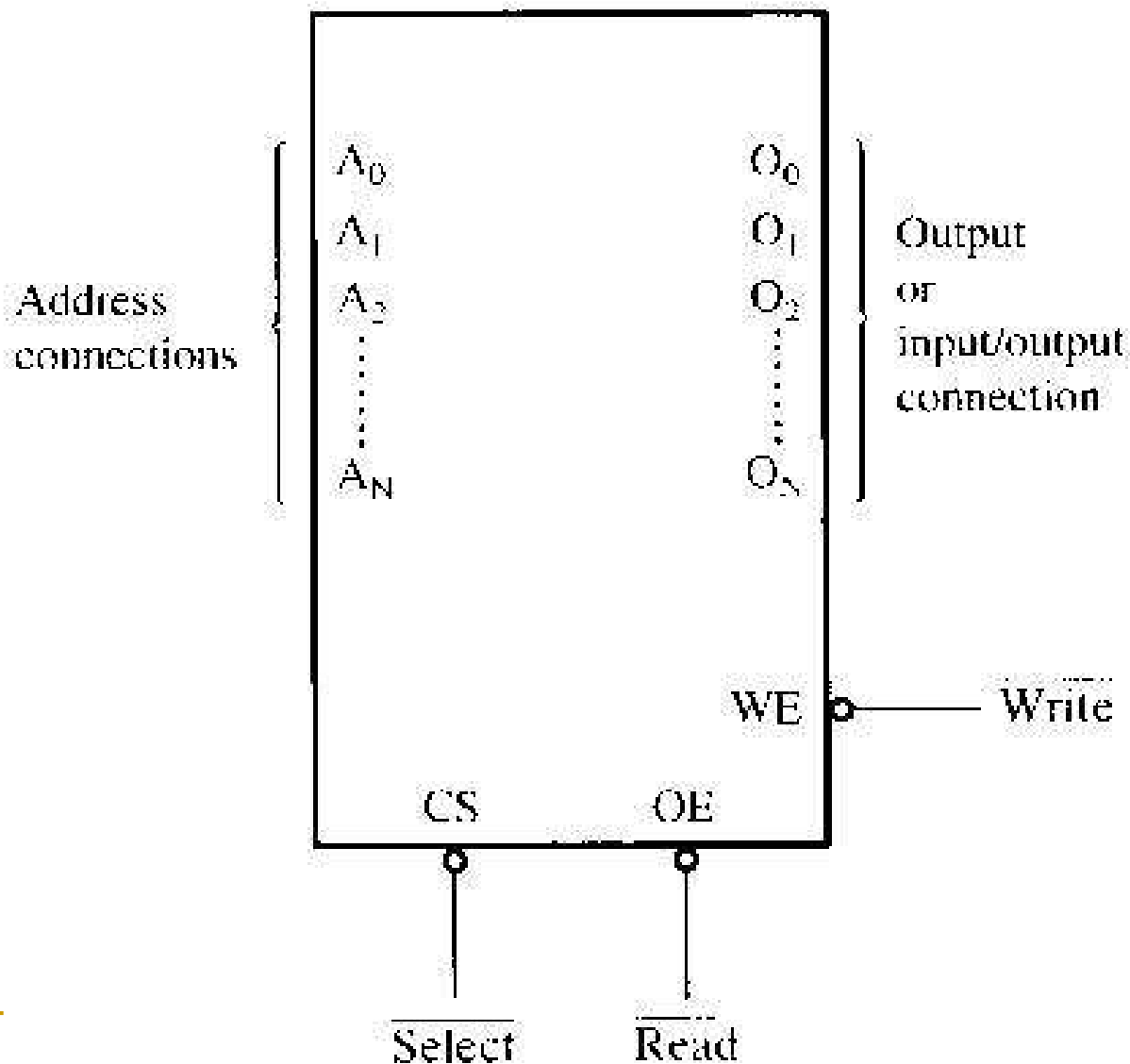
■ Memory Speed

- Access Time

Pin configuration of Memory Chips

- Pin connections common to all memory devices.
 - The address inputs
 - Data outputs or input
 - Some type of selection input
 - At least one control input used to select a read or write operation.
- Memory device has one chip select, chip enable or select
- Memory device has control pin

Pin configuration of Memory Chips



Address Connections

- Memory devices have address inputs to select a memory location within the device.
- The number of address pins on a memory device is determined by the number of memory locations found within it.
- A 1K memory device has 10 address pins. therefore, 10 address inputs are required to select any of its 1024 memory locations.
- It takes a 10-bit binary number to select any single location

Data Connections

- All memory devices have a set of data outputs or input/outputs.
- An 8-bit-wide memory device is often called a **byte-wide** memory.
- Catalog listings of memory devices often refer to memory locations times bits per location.
 - ❑ A memory device with 1K memory locations and 8 bits in each location is often listed as a 1K x 8 by the manufacturer
 - ❑ Memory devices are often classified according to total bit capacity.

Select Connections

- Each memory device has an input that selects or enables the memory device.
- This type of input is most often called a **chip select** (G2A) **chip enable** (CE) or simply **select** (S) input.
- RAM memory generally has at least one or input, and ROM has at least one
- If more than one CE connection is present, all must be activated to read or write data.

Control Connections

- All memory devices have some form of control input or inputs.
 - ROM usually has one control input, while RAM often has one or two control inputs
- Control input often found on ROM is the **output enable** or **gate** connection, which allows data flow from output data pins.
- RAM has either one or two control inputs.
 - if one control input, it is often called R/W

Control Connections

- If the RAM has two control inputs, they are usually labeled WE (or W), and OE (or G).
 - **write enable** must be active to perform memory write, and OE active to perform a memory read
 - when the two controls are present, they must never both be active at the same time
 - If both inputs are inactive, data are neither written nor read.

Memory Hierarchy

- ❑ Assume a memory 16 X 8
- ❑ Address Pins required $2^4 = 16$
- ❑ So 4 Address pins are required i-e: A0 to A3
- ❑ Data pins, always D0 to D7

Address		Actual Memory Location							
000F									
000E									
000D									
000C									
000B									
000A									
0009									
0008									
0007									
0006									
0005									
0004									
0003									
0002									
0001									
0000									

ROM Memory

- Read-only memory (ROM) permanently stores programs/data resident to the system.
 - ❑ must not change when power disconnected
 - ❑ Often called nonvolatile memory, because its contents do not change even if power is disconnected.
 - ❑ programmed during fabrication at the factory
- The EPROM (erasable programmable read-only memory) is commonly used when software must be changed often.
 - ❑ An EPROM is programmed in the field

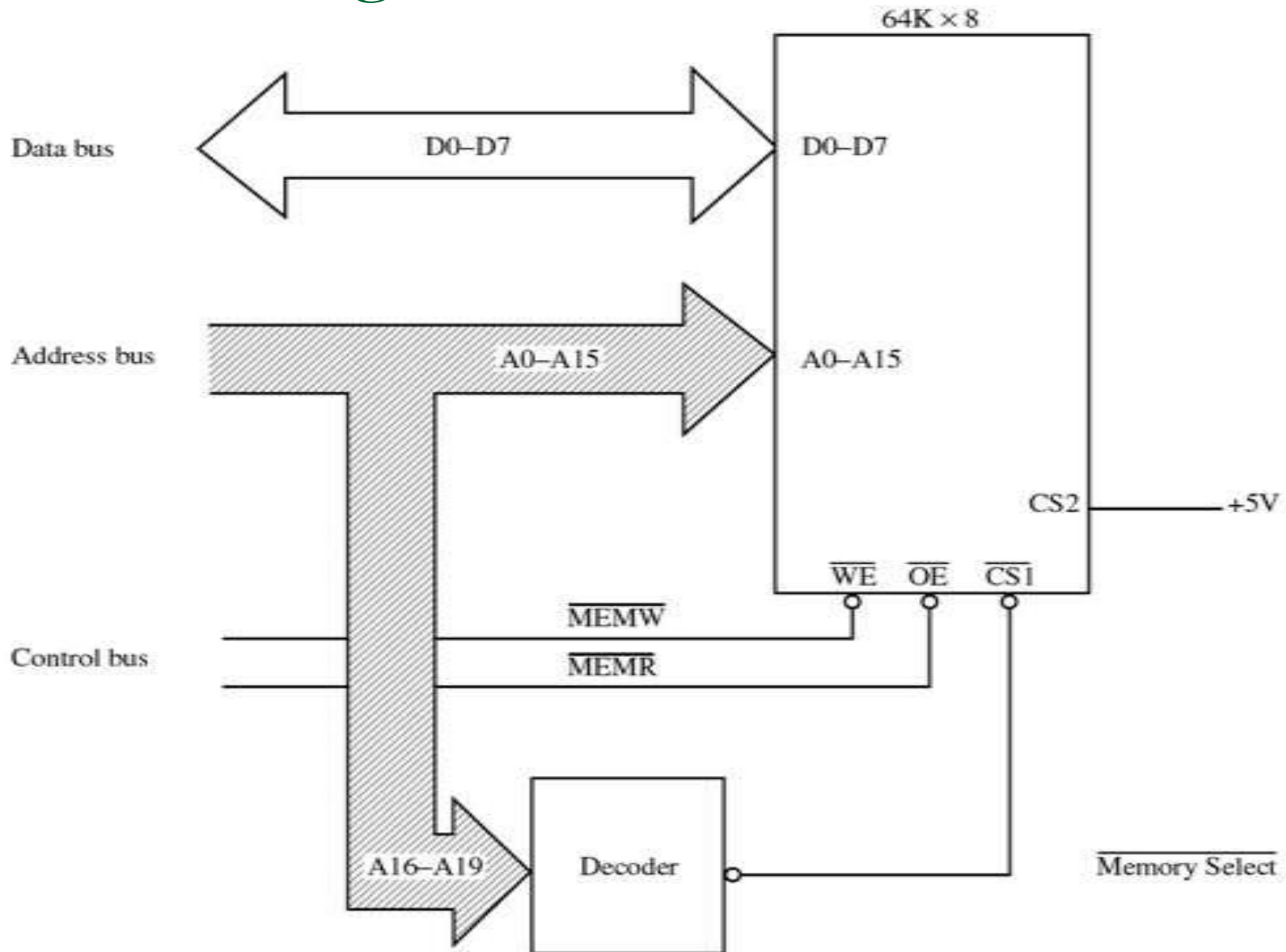
ROM Memory

- Type of EPROM
 - PROM memory devices are also available, although they are not as common today.
- The PROM (programmable read-only memory) is also programmed in the field
 - Once it is programmed, it cannot be erased.
- A newer type of read-mostly memory (RMM) is called the flash memory.
 - also often called an EEPROM (electrically erasable programmable ROM), EAROM (electrically alterable ROM) or NOVRAM (nonvolatile RAM)

ROM Memory

- ❑ Electrically erasable in the system, but they require more time to erase than normal RAM.
- ❑ The flash memory device is used to store setup information for systems such as the video card in the computer.
- ❑ Flash has all but replaced the EPROM in most computer systems for the BIOS.
- ❑ some systems contain a password stored in the flash memory device

Interfacing with 8088



Why Decode Memory?

- The 8088 has 20 address connections and the **RAM/ROM** has X connections.
- The 8088 sends out a 20-bit memory address whenever it reads or writes data.
- – because the **RAM/ROM** has only X address pins, there is a mismatch that must be corrected
- The decoder corrects the mismatch by decoding address pins that do not connect to the memory component.
- Types of Decoder
 - NAND Gate Decoder
 - 3-8 Line Decoder

NAND Gate Decoder

- When the 2K x 8 SRAM is used:
 - ❑ 2K X 8 EPROM, $2^{11} = 2K$
 - ❑ Address connections $A_{10}-A_0$
 - ❑ Remaining 9-pins ($A_{19}-A_{11}$) are used for Decoding

Example

A 2K EPROM is decoded at memory address locations FF800H

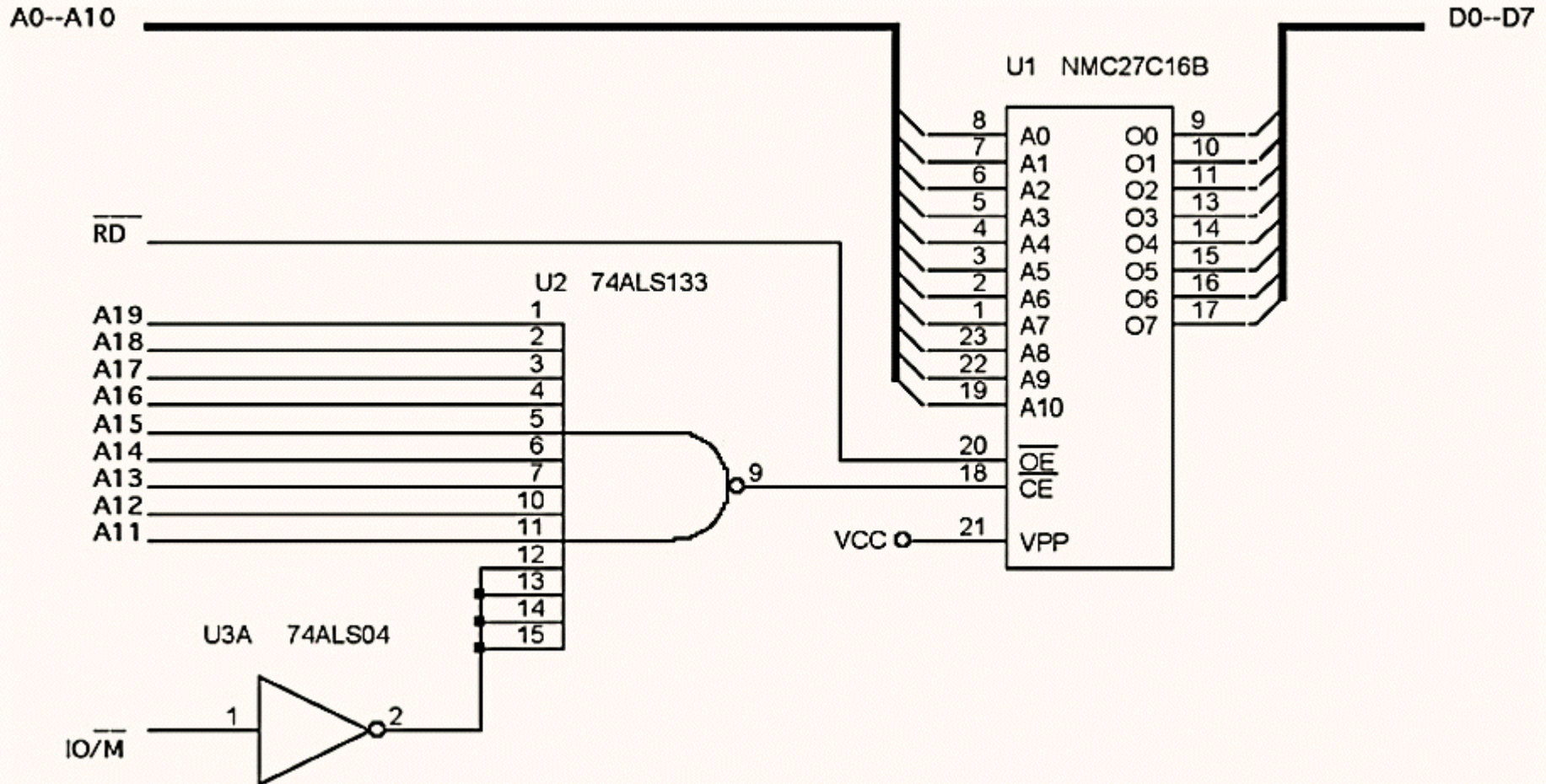
Starting Address = 1111 1111 1000 0000 0000 = FF800H

Ending Address = FF800H + 2K -1 = FFFFFH

1111 1111 1111 1111 1111 = FFFFFH

1111 1111 1XXX XXXX XXXX

NAND Gate Decoder

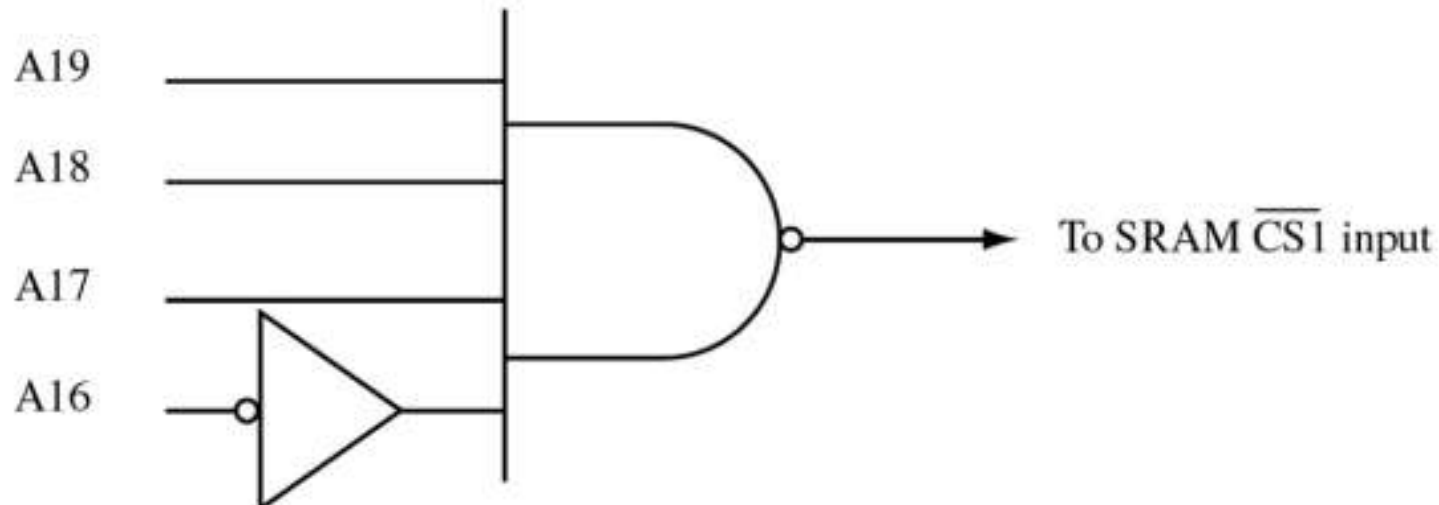


- Address FF800H to FFFFFH.

NAND Gate Decoder

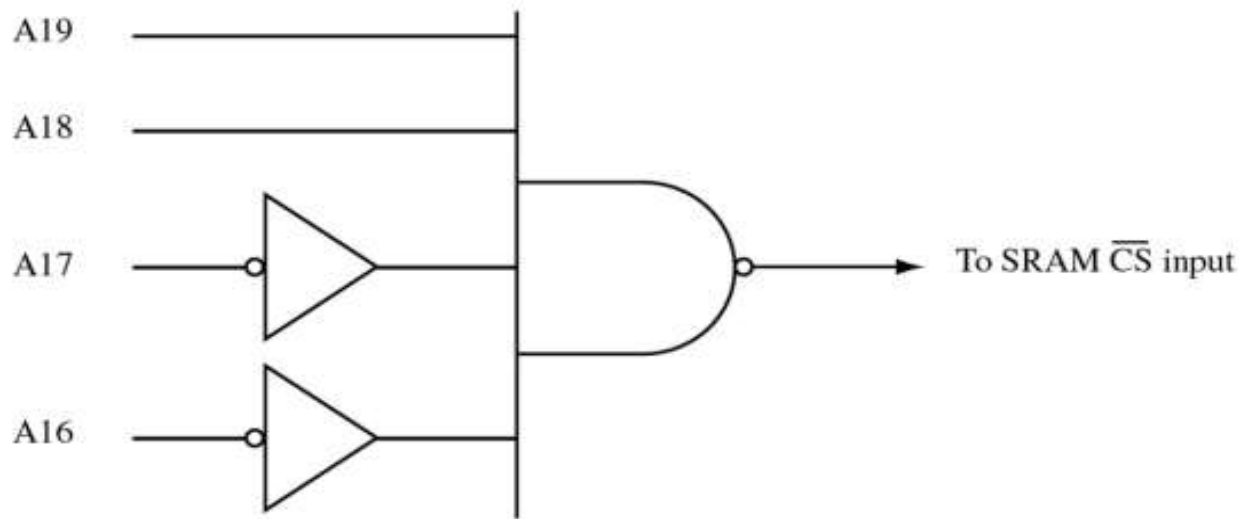
- When the 64K x 8 SRAM is used:
 - ❑ Address connections A15–A0 of 8088 are connected to address inputs A15–A0 of the SRAM.
 - ❑ The remaining four address pins (A19–A16) are connected to a NAND gate decoder

NAND Gate Decoder



- In this circuit a NAND gate decodes the memory address, as seen in the above figure.
- The memory will be enabled only when $A19-A16 = 1110$.
- The 64K SRAM is mapped to the address range $E0000H$ to $FFFFFFH$.

NAND Gate Decoder



- In this circuit a NAND gate decodes the memory address, as seen in the above figure.
- The memory will be enabled only when $A19-A16 = 1100$.
- The 64K SRAM is mapped to the address range $C0000H$ to $CFFFFH$.

NAND Gate Decoder

A circuit containing 32KB of RAM is to be interfaced to an 8088 based system, so that the first address of the RAM is at 48000H. What is the entire range of the RAM Address? How is the address bus used to enable the RAMs? What address lines should be used?

4	8 → F	0 → F	0 → F	0 → F
A_{19} A_{18} A_{17} A_{16}	A_{15} A_{14} A_{13} A_{12}	A_{11} A_{10} A_9 A_8	A_7 A_6 A_5 A_4	A_3 A_2 A_1 A_0
0 1 0 0	1 X X X	X X X X	X X X X	X X X X